Who can relate to Hamid Sourian after midterm exam?
Is one test enough to pass/fail?
Who wants a second test?



Prof Fani is cool 💪

I found one of the questions on the exam to be a little confusing (the semantics one: I wasn't sure what exactly it was asking of me) but other than that I believe the test was fair. So far I have found it difficult to try and relate what my group is doing for our project and what we are being taught in lecture but I understand it won't be a direct correlation.

The way Hoseeins is teaching with real examples and trying to connect subject in proper order is good. The only thing is he can make the things more clear like interms of expectations , for example in midterm Question 2 and Question 3 literally unable to understand with the wording of those.

Provide practice exams beforehand please

would like a practice exam to be posted for midterms

After reviewing the midterm answer key, I feel pretty confused by the questions. I did not understand what q3 or q4 were asking about. For q3, I felt that the question was asking about how words would be positioned relative to one another in Euclidean space based on onomatopoeia, rather than how to construct the underlying representation using a letter-frequency matric. I also felt like q4 was unclear as it asked specifically if precision or recall is better for the problem. Which seems like a comparison of two things rather than comparing all four possible scenarios. The lectures and the constant 10/10 media references are very engaging. I also appreciate how they are tied to real world events and are recorded so we can view them asynchronously. However, I felt that some of the exam questions were worded in a way that didn't clearly communicate what the expected answers were, and I think I could have answered them better if I had understood the intended interpretation of the questions.

I believe that although the questions themselves were not overly difficult on the midterm, I would like some more clarification on types of questions asked (such as specifying definitions were something we should have memorized). I do think that a fair amount of information was revealed, but I found myself focusing heavily on topics that ended up being redundant for the test. While this is most definitely a fault of my own, I would not be surprised if there was a number of students in the same boat. So if I had a solution, I would request that a practice midterm is distributed for subsequent midterms. But I don't think any more information on what the questions are involving is mandatory

The midterm question wording was not super clean on what you wanted, so I wrote some answer that may not be satisfactory. Additionally no practice midterm extremely harms success in exams, without ideas of what you are looking for when answering question, studying becomes less clear on what to focus on, and no frame of reference on style of answers.

In my opinion, we spend a bit too much lecture time recapping the previous lecture. Also and relatedly, going overtime on a regular basis is not polite: some students only have 10 minutes between classes, and may need to walk across campus and/or use the bathroom.

Really had a bad exam

I've really enjoyed this class so far! My only critique might be that in some lectures certain topics (like the confusion matrices) are a bit over explained (maybe it's just the nature of formalizing these things) but in general they're pretty clear.

I honestly thought that another option could be added for "How's Hossein's teaching?" which is that "Great, I really enjoy the class" because I really do enjoy coming to SENG474/ CSC503. I really appreciate Prof's teaching, especially how he spends time to go through an review what was previously covered briefly, as it often helps ground me/ remind me what we went through and also for me to revisit any weaker concepts/ knowledge.

I really appreciate the pace of the class as well. I feel that Data Mining/ Machine Learning can be very complicated and hard to digest but this pace makes it much more breathable without overwhelming us with content. I have also received a lot of help from Prof when it comes to sending him emails outside of class, asking him questions after class and Prof has been really helpful and friendly. I really appreciate his patience and understanding.

Thank you so much Prof! Fret not, I did not learn to just not be a brick in the wall, I have really learnt a lot more and enjoy the lectures you have provided.

please provide sample exams and please make the exam wording less confusing.

Overall, I think the course is going well and I'm happy with it.
The only thing I would change is the class duration. As you said yourself, a 50-minute exam is "just the blink of an eye," and honestly, so is a 50-minute lecture. Around 1.5 hours per session would give us more time to explore the topics in depth.

The midterm was really hard, also the questions seemed too confusing specially the Q2 and Q3.

My critique is that other professors don't practice such masterful pedagogy as Hossein! Seriously, Hossein is like Steve Jobs, mixing the technical and non-technical, the science and the liberal arts, so as to create a more palatable and fruitful learning experience for us. I probably will create a rate my professor account just so I can give him a 5/5.

Give practice questions from the optional textbook to go with the lectures. I like doing practice questions.

Project could use a bit more structure and guidance on picking a topic

I'm using this space for comments. I think you're a great teacher and I like that you apply concepts to real applications that have enormous impacts. I think the NLP embedding is a bit long. I didn't like question 3 on the midterm at all. I didn't understand it and my test score is going to suffer because of it. Once an explanation was given during the test for questions 3 and 4, it made more sense but I think many students would have otherwise answered not enough or not hit exact points without explicit instruction. I put "expected easier exam" above but only due to lack of choices. Really, it was just question 3 and I'm honestly still confused about what it was asking. Otherwise, I didn't expect an easier exam and thought it was good.

Lectures have been engaging so far, course work is also really interesting! Only critiques currently would be requesting practice midterms/example questions be provided for the midterm/exams before hand.

I expected there to be a sample question bank or a better description of what kind of problems we were gonna be asked on the midterm. 1 exam (gave us a better idea of how to prepare). I was confused whether to read more theory or math for this midterm, also the questions were unexpectedly difficult on the first midterm exam.

I felt that we did not have enough study resources for the exam. I would appreciate having a practice exam or something similar with answer keys.

Test was not at all what I expected. Would love a bit more pre-context as to what expect. I studied concepts and practiced vigously. My take is that the questions were quite vague and difficult to understand what they are asking for. I understand all the concepts that they were asking well, but I struggled understanding what they were looking for as answers. In the future id recommend more straight forward questions, maybe some more technical stuff. it was a bit vague and difficult to comprehend.

I found the exam very ambiguous and spent a large amount of exam time trying to figure out what question meant and what format it the answer should be in. There was also not enough time and it was a lot more challenging than I expected. In the future, I would appreciate more explicit instructions, such as what you are looking for in an answer.

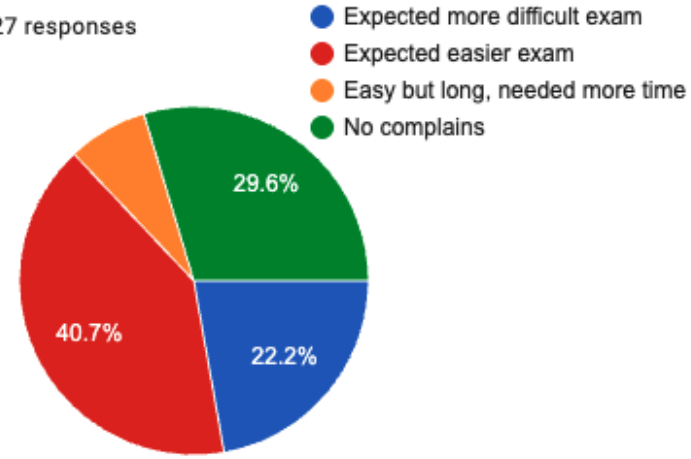
Have some multiple choice questions would be better, 5 written questions are not bad but my hand feels sore lol

lectures could be more organized

Examples given are very pessimistic, and formulas are not explained very well in class.

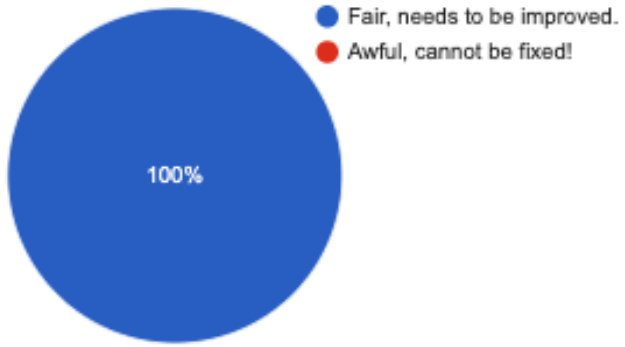
How was the midterm exam?

27 responses



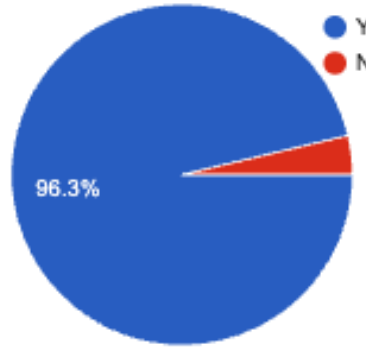
How's Hossein's teaching?

27 responses

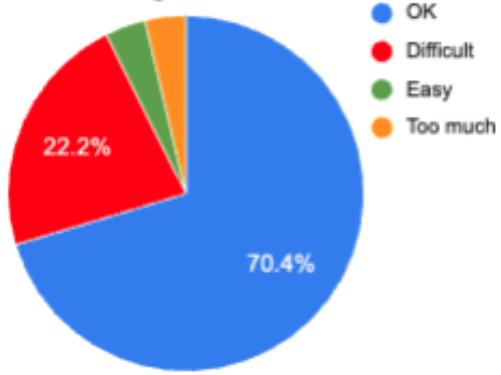


Is it worth doing all this coursework:

27 responses

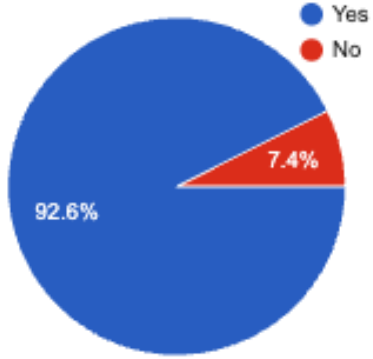


Is it worth taking this course?



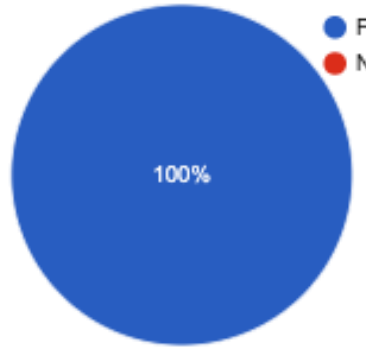
Is it worth taking this course?

27 responses



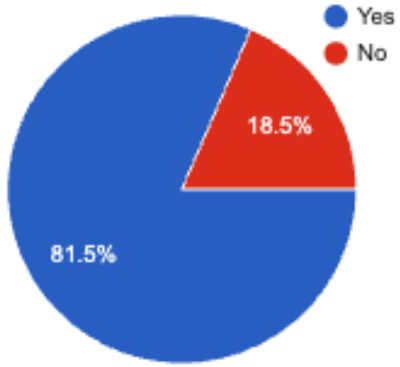
How's marking?

27 responses

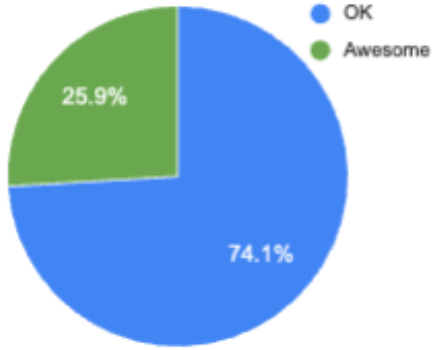


Interested in screening Blade Runner (1982, Ridley Scott)?

27 responses

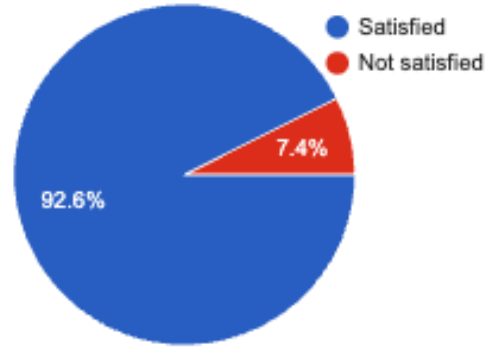


Is it worth taking this course?



How are TAs?

27 responses



Outline: W06C

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k-Nearest Neighbor

IIII) Evaluation

Metrics, Methodology: How to Run Evaluation?

(1 Final) vs. (2 Midterms + 1 Final)

5

Evaluation Methodology: Protocol to Count Inferences

Contingency Table, Confusion Matrix

		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error
	Predicted condition negative	**False negative**, Type II error	**True negative**

What would be the observations (questions)

On which the model (student) make inferences (answers)

About their labels (true answers)…

Evaluation Methodology: Protocol to Count Inferences

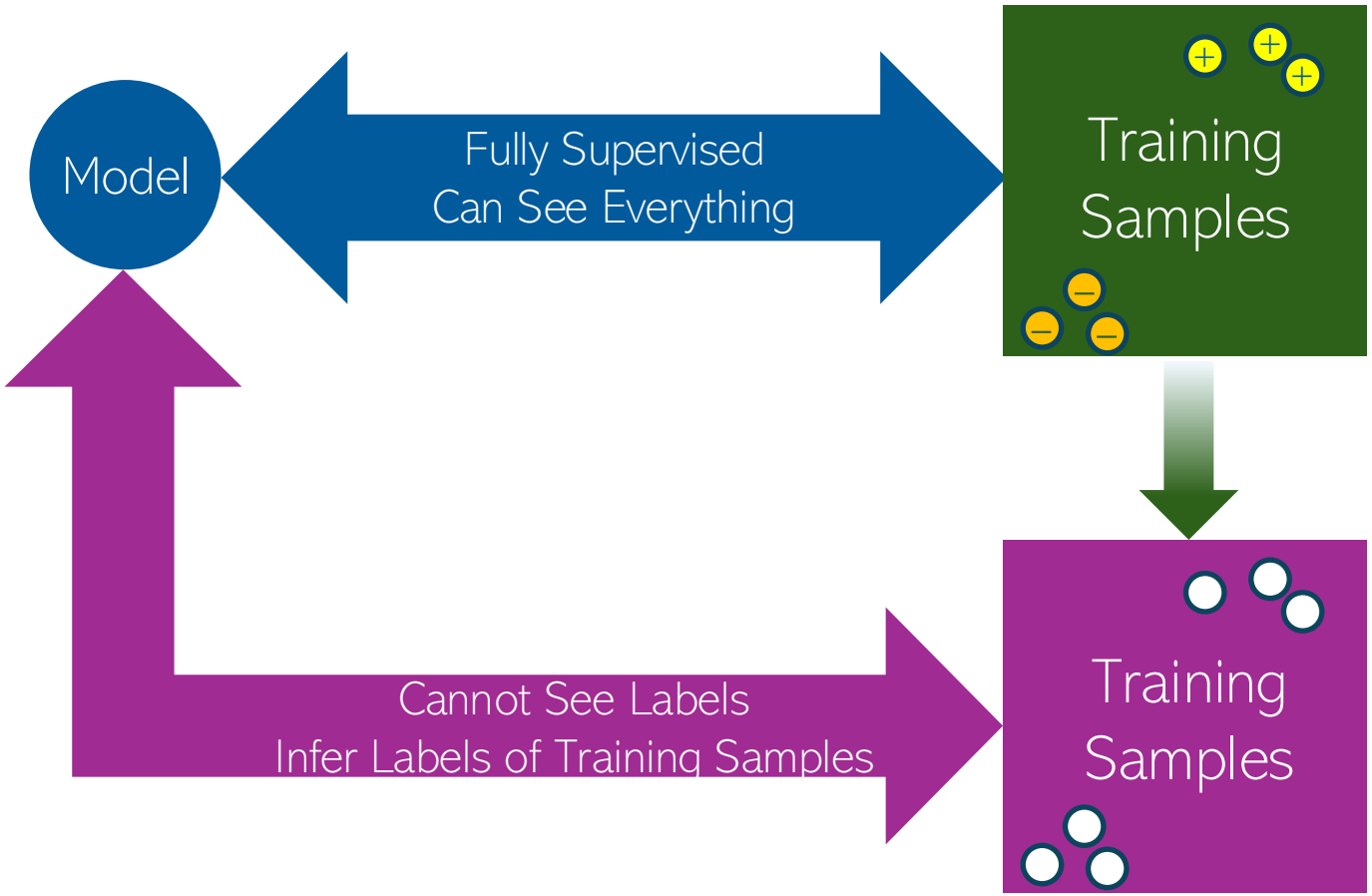


Students have the questions and answers in the slides.
Now, evaluate their performance on what questions?

Evaluation Methodology: Protocol to Count Inferences

Evaluate on the **same training set**

Simple, but time consuming (training set may be large)

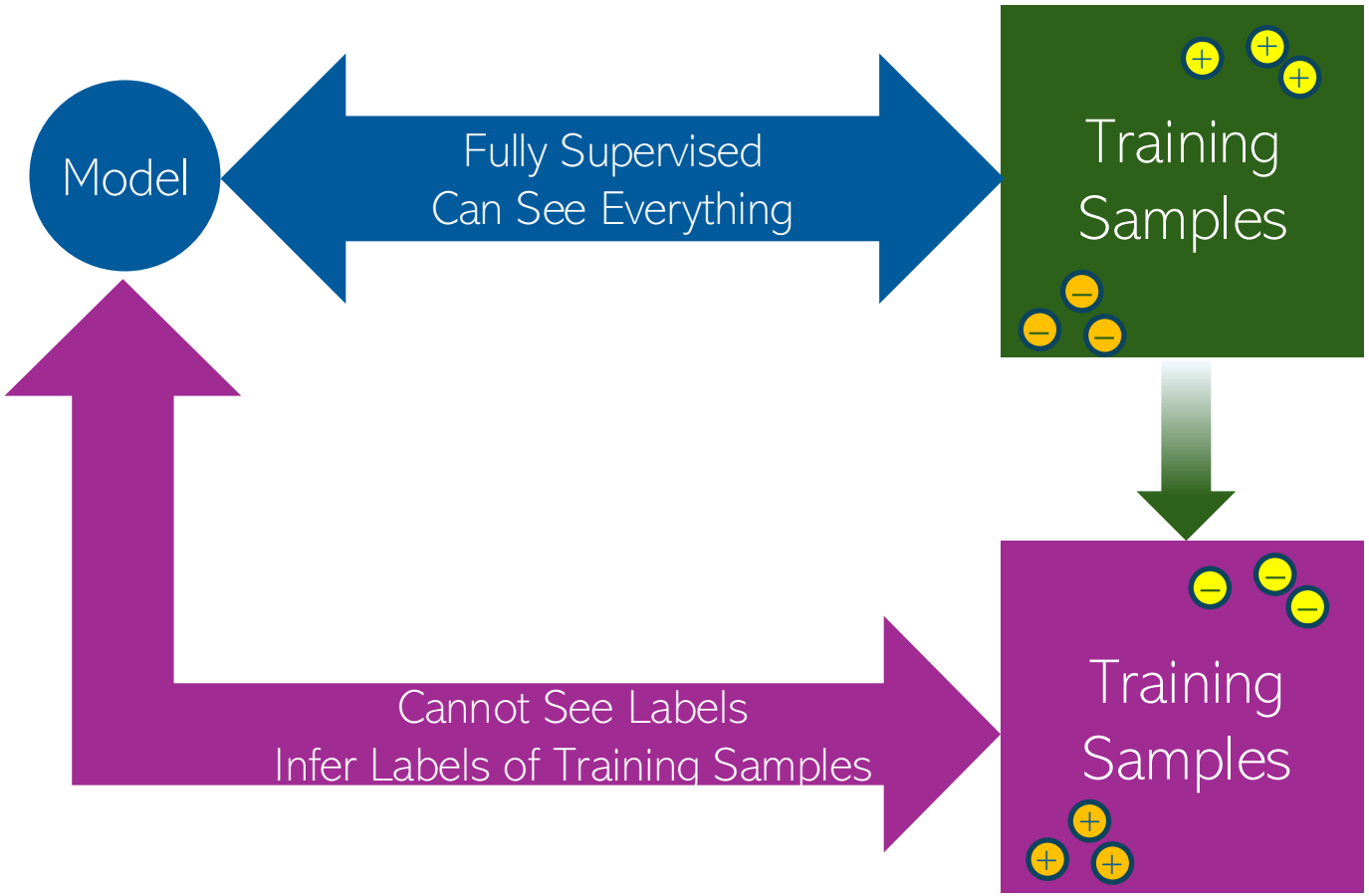


		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error
	Predicted condition negative	**False negative**, Type II error	**True negative**

Evaluation Methodology: Protocol to Count Inferences

Evaluate on the **same training set**

Simple, but time consuming (training set may be large)



Underfitting

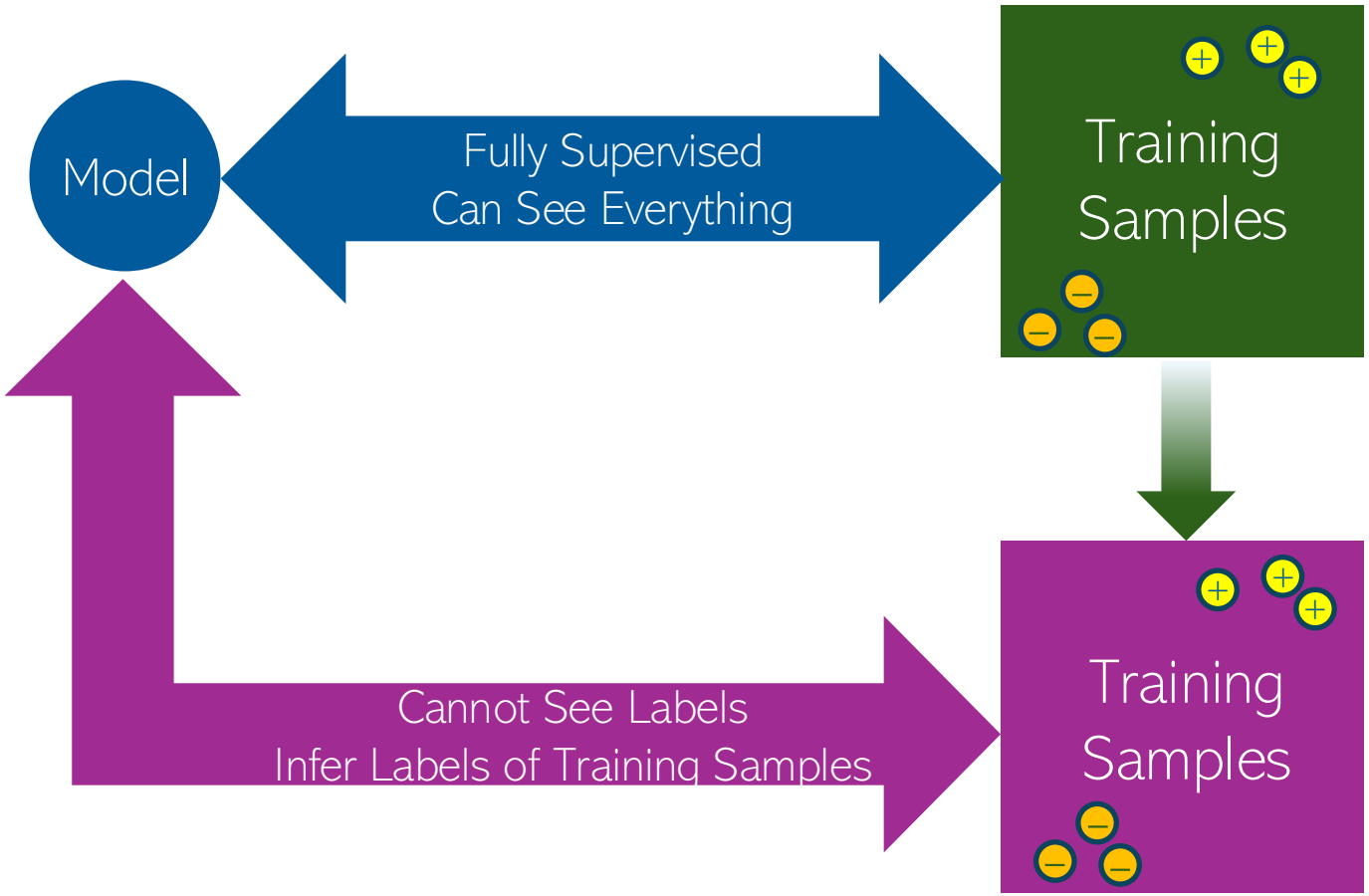
The model cannot learn!

		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	0 **True positive**	3 **False positive**, Type I error
	Predicted condition negative	3 **False negative**, Type II error	0 **True negative**

Evaluation Methodology: Protocol to Count Inferences

Evaluate on the **same training set**

Simple, but time consuming (training set may be large)



The model can learn.

An Oracle? Of course not!

It already knew the labels. No magic ...

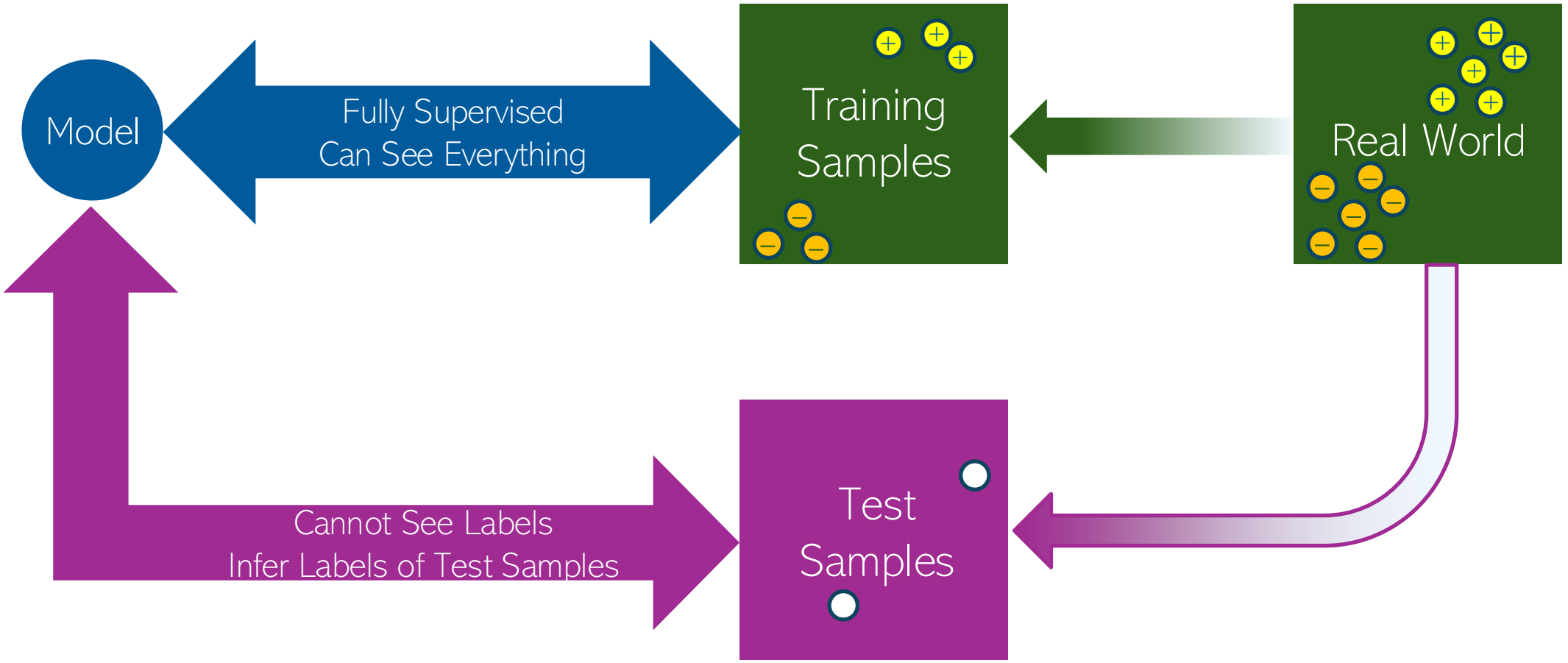
		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	3 **True positive**	0 **False positive**, Type I error
	Predicted condition negative	0 **False negative**, Type II error	3 **True negative**



Evaluation Methodology: Protocol to Count Inferences

Evaluate on a different **unseen samples: test set**

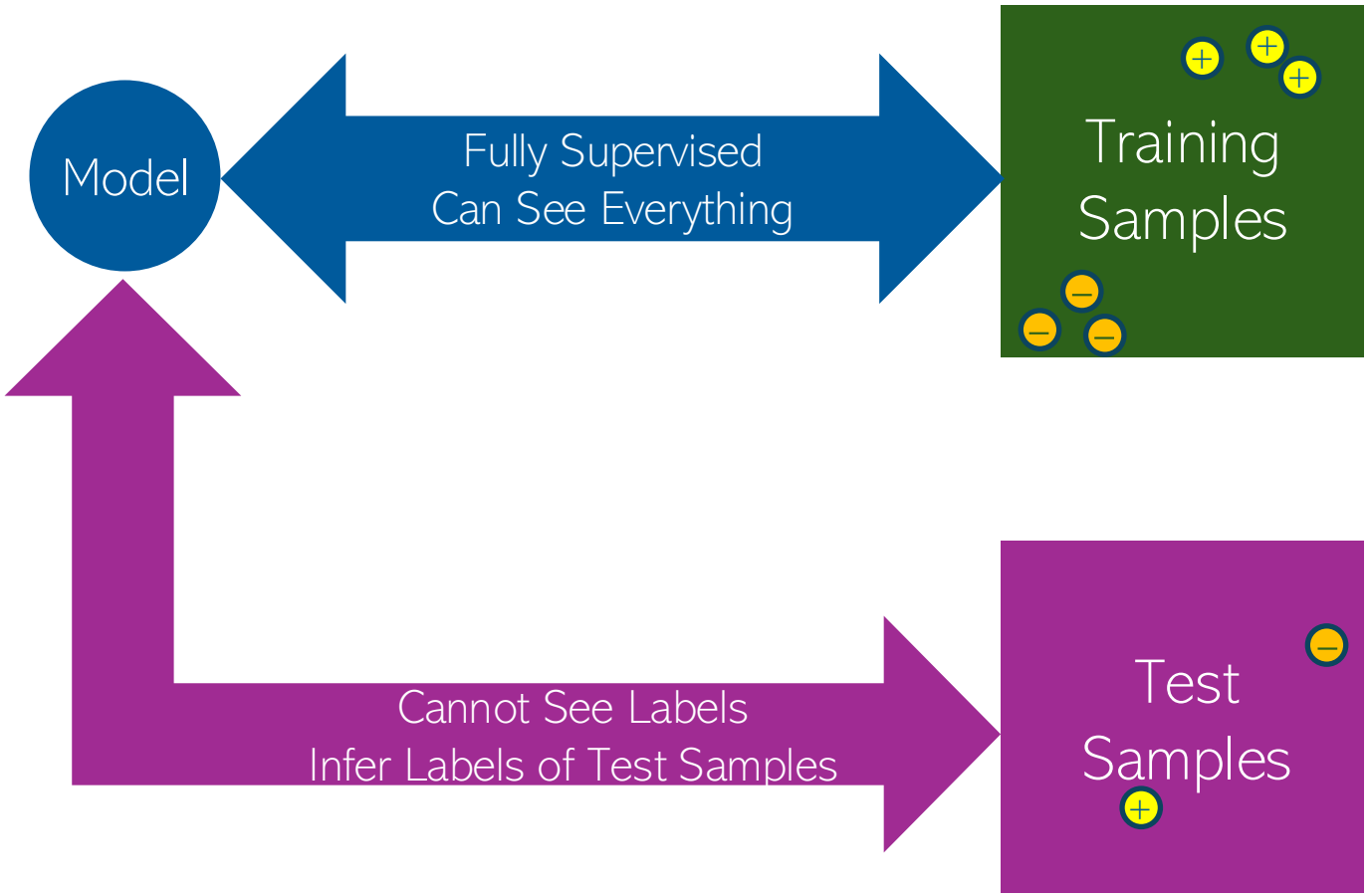
Fast (test set is commonly small), but very challenging



Evaluation Methodology: Protocol to Count Inferences

Evaluate on a different **unseen samples: test set**

Fast (test set is commonly small), but very challenging



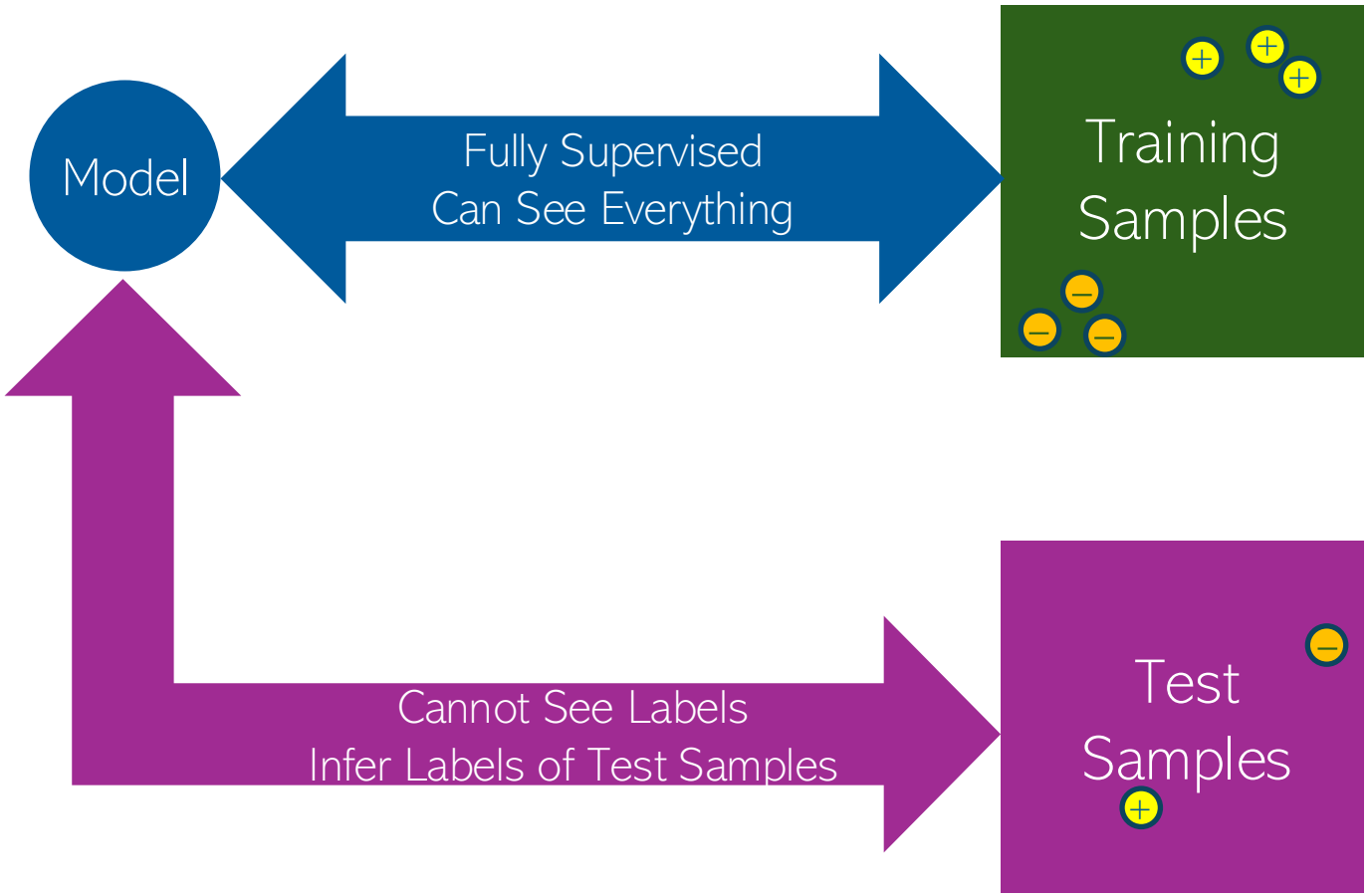
Poor Performance. Why?

		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	0 **True positive**	1 **False positive**, Type I error
	Predicted condition negative	1 **False negative**, Type II error	0 **True negative**

Evaluation Methodology: Protocol to Count Inferences

Evaluate on a different **unseen samples: test set**

Fast (test set is commonly small), but very challenging



		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	0 **True positive**	3 **False positive**, Type I error
	Predicted condition negative	3 **False negative**, Type II error	0 **True negative**

Underfitting Model. Expected.

We should fix the model.

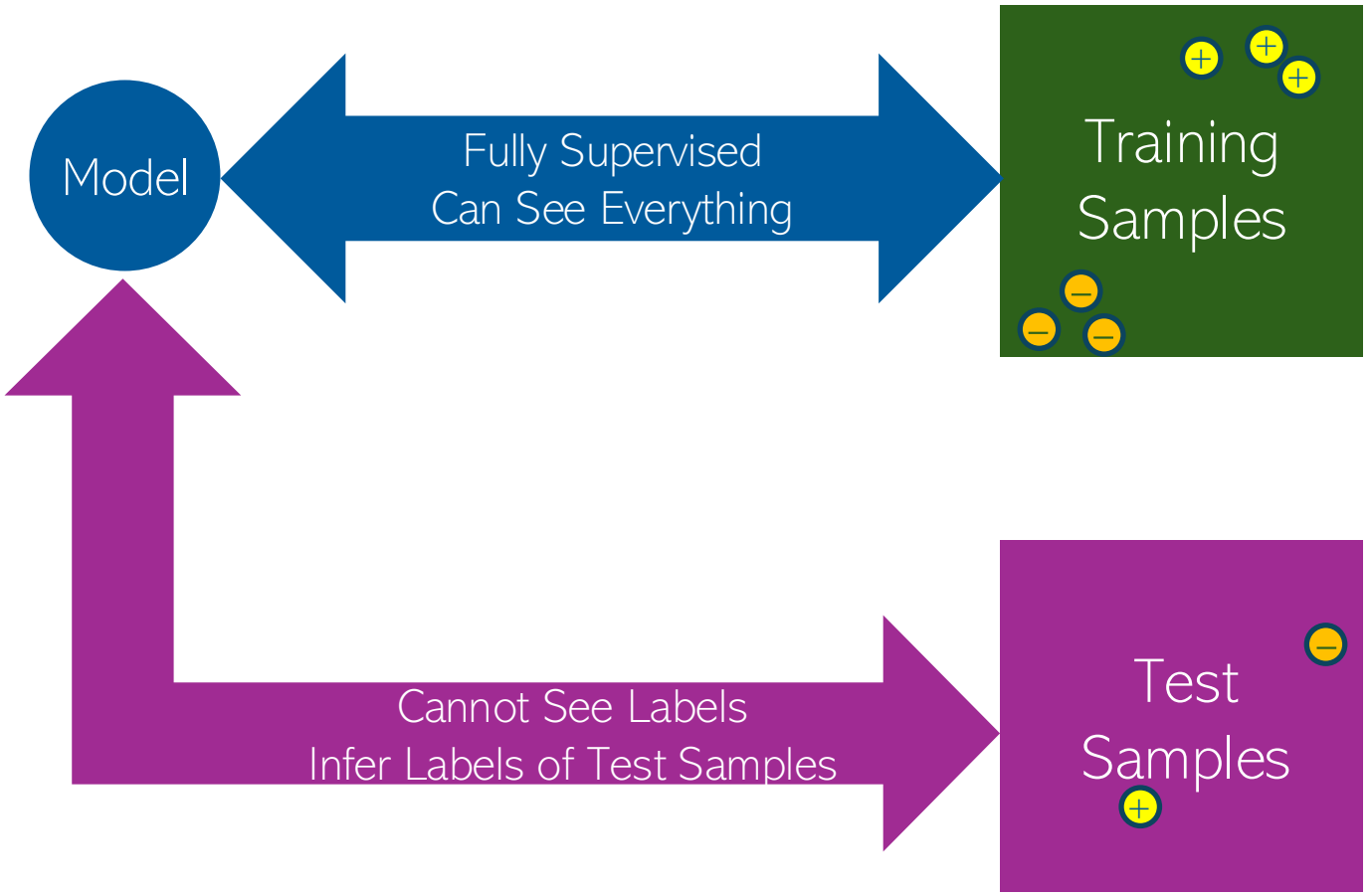
		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	0 **True positive**	1 **False positive**, Type I error
	Predicted condition negative	1 **False negative**, Type II error	0 **True negative**

14

Evaluation Methodology: Protocol to Count Inferences

Evaluate on a different **unseen samples: test set**

Fast (test set is commonly small), but very challenging



		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	3 **True positive**	0 **False positive**, Type I error
	Predicted condition negative	0 **False negative**, Type II error	3 **True negative**

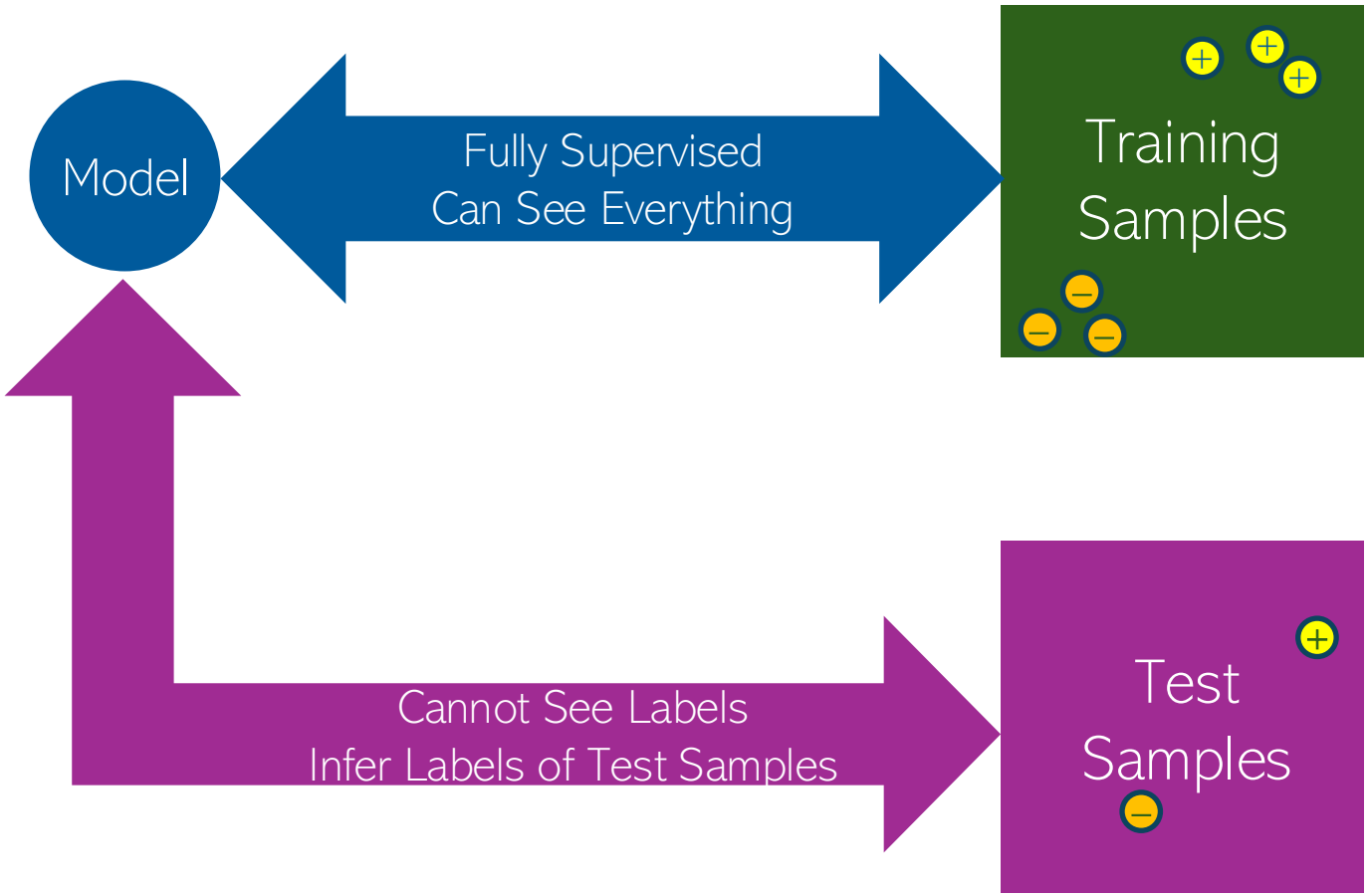
Overfitting to Training Set.
The model cannot generalize.

		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	0 **True positive**	1 **False positive**, Type I error
	Predicted condition negative	1 **False negative**, Type II error	0 **True negative**

Evaluation Methodology: Protocol to Count Inferences

Evaluate on a different **unseen samples: test set**

Fast (test set is commonly small), but very challenging



An Oracle.

No need to evaluate on train set.

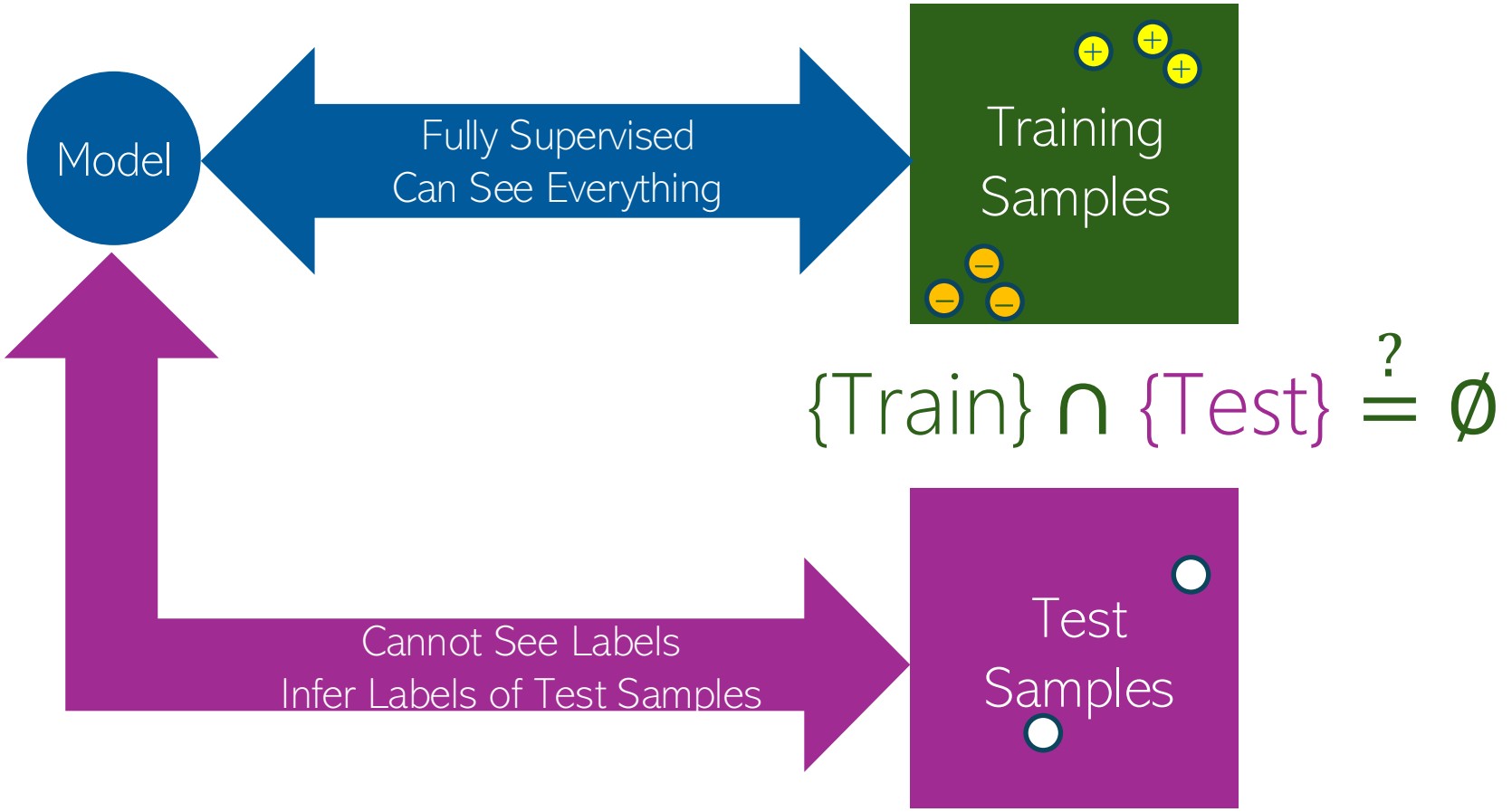
Rarely happens!

		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	1 **True positive**	0 **False positive**, Type I error
	Predicted condition negative	0 **False negative**, Type II error	1 **True negative**

Evaluation Methodology: Protocol to Count Inferences

Should training and test sets be **disjoint**, or an **overlap** acceptable?

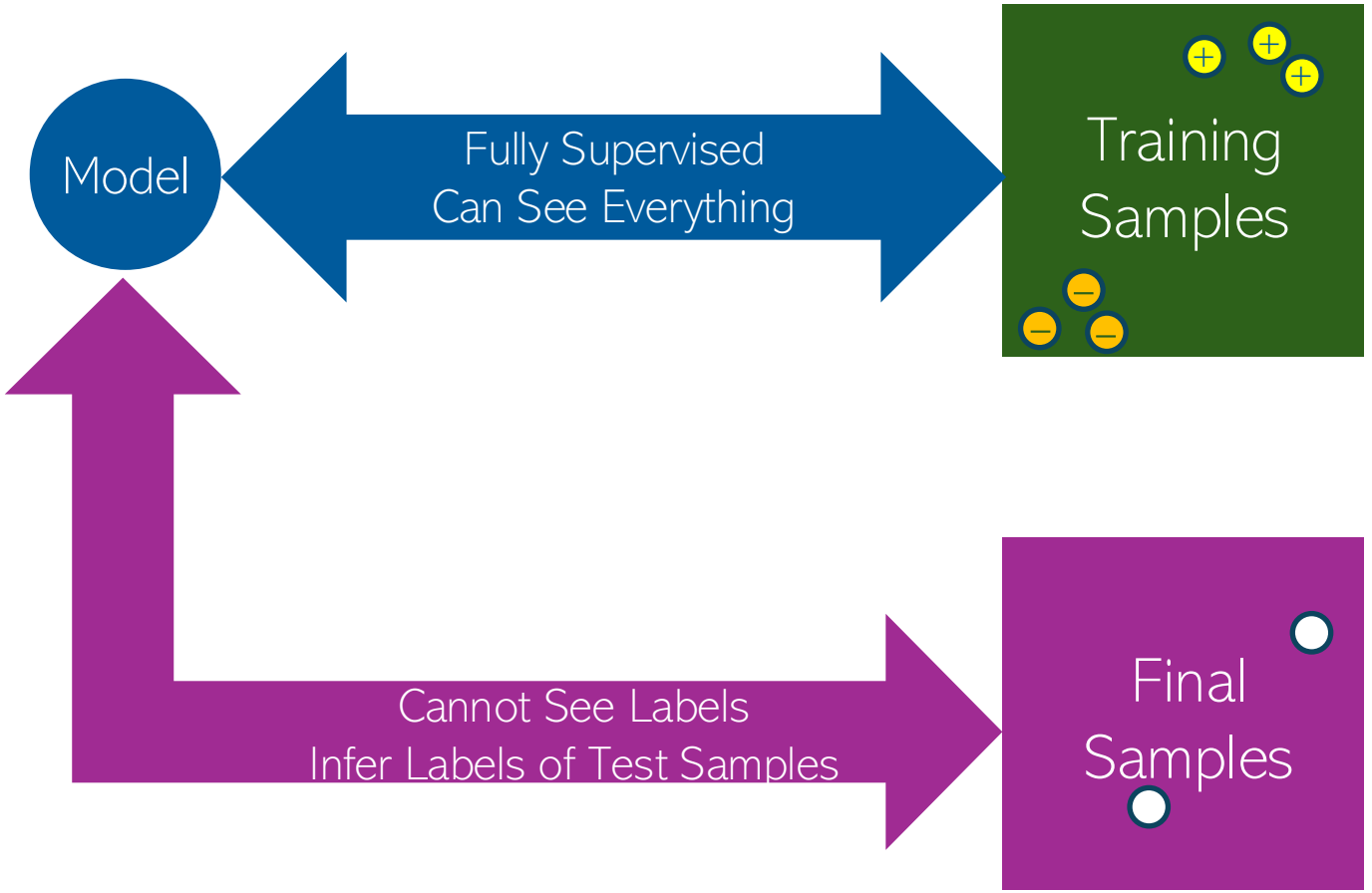
Think of students in an exam. Should all questions be unseen before?



Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a **single test set** enough to decide on a model?

*Think of students in **only one exam** (final exam). Is it sufficient to pass/fail?*

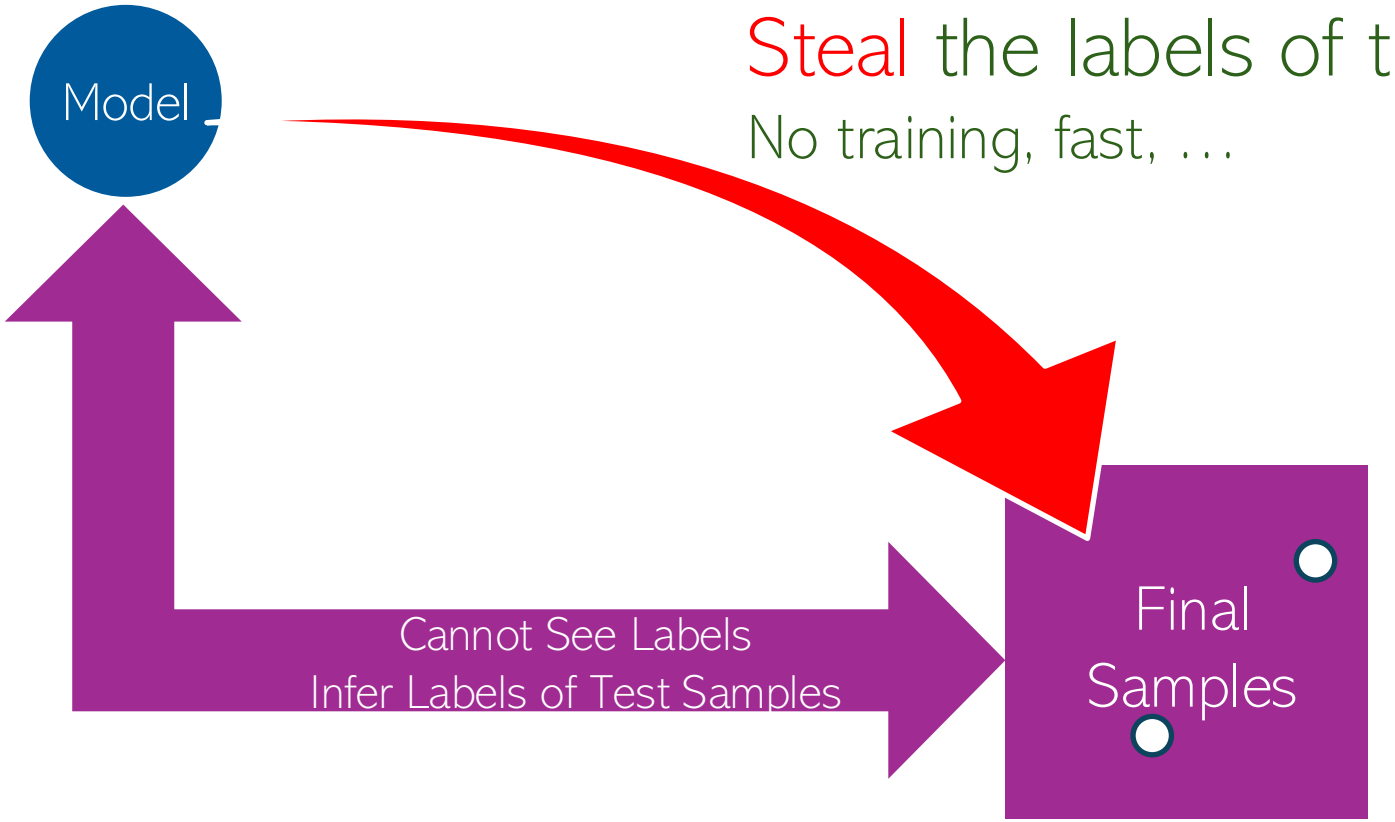


Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a **single test set** enough to decide on a model?

*Think of students in **only one exam** (final exam). Is it sufficient to pass/fail?*

Yes.







Mongrel Jedi

PetFinder.my Contest: 1st Place Winner Disqualified

307

Posted in petfinder-adoption-prediction a year ago

Dear Participants,

We would like to announce that the 1st Place Team, Bestpetting has been disqualified from the contest for cheating. The Kaggle Grandmaster cheater has also been permanently banned on this platform as the evidence points towards him being the key party behind this fraudulent activity.

Here is what the Bestpetting team did in the PetFinder.my contest:

- They fraudulently obtained adoption speed answers for the private test data (possibly by scraping our website)

Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a **single test set** enough to decide on a model?

*Think of students in **only one exam** (final exam). Is it sufficient to pass/fail?*

Yes.



Obtain the labels of the test set!

No training on the train set but on evaluation results!

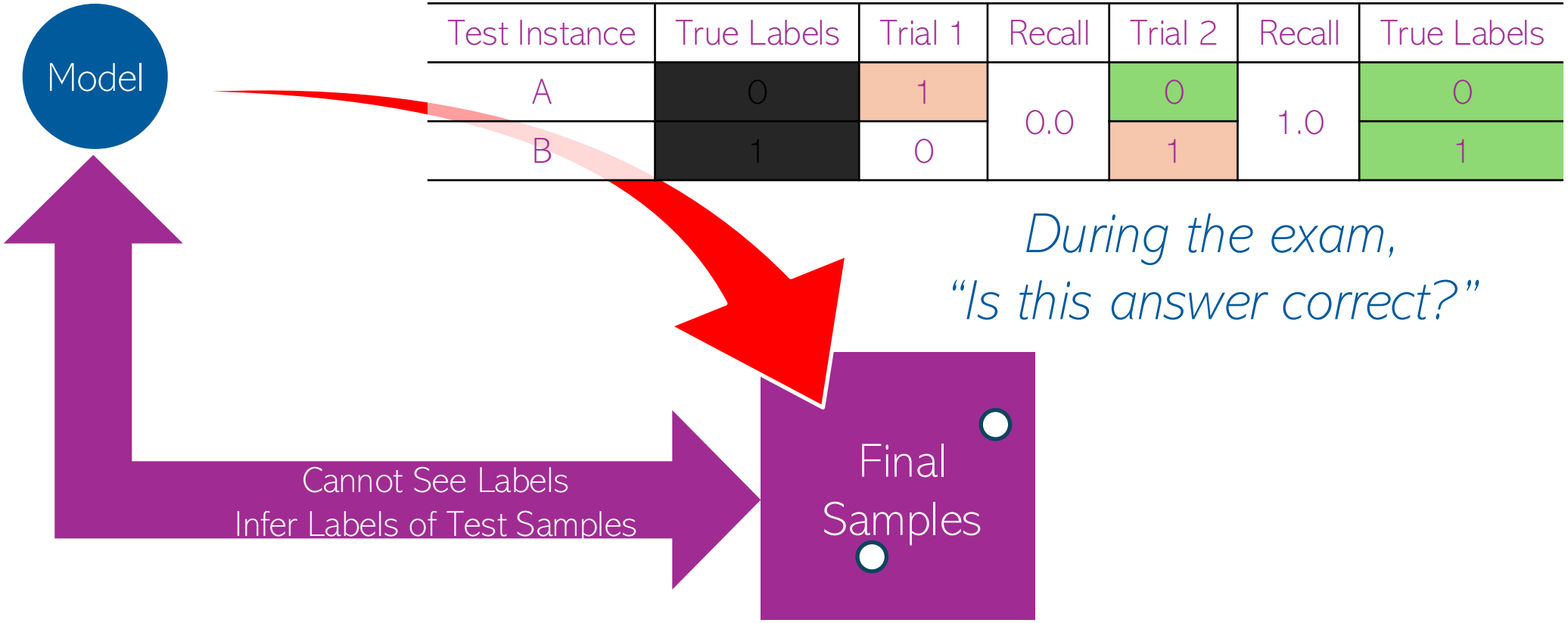
Learning the test set's labels.

Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a **single test set** enough to decide on a model?

*Think of students in **only one exam** (final exam). Is it sufficient to pass/fail?*

Yes.



		Submissions				
Test Instance	True Labels	Trial 1	Recall	Trial 2	Recall	True Labels
A	0	1	0.0	0	1.0	0
B	1	0		1		1

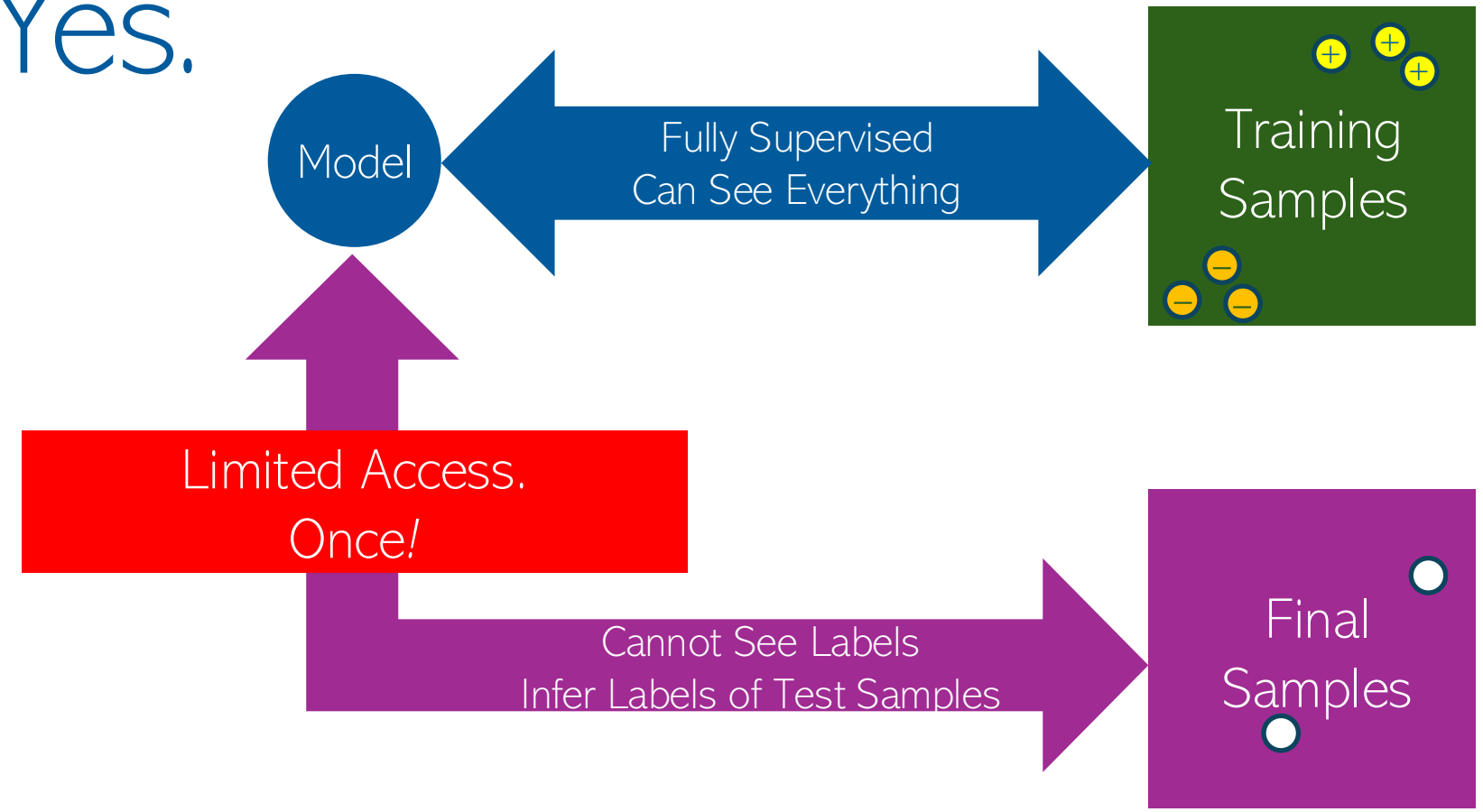
During the exam,
"Is this answer correct?"

Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a **single test set** enough to decide on a model?

*Think of students in **only one exam** (final exam). Is it sufficient to pass/fail?*

Yes.



Evaluation Methodology: Protocol to Count Inferences

A **single test set**

- Labels are protected from stealing
- Access to labels is limited (once) to block learning labels by trial-errors

Is it fair for the models (*students*)?

Is it sufficient for the developer (*instructor/school*)?

What were your answers before and after the midterm exam?



Medal record			[hide]
Representing Iran			
Men's Greco-Roman Wrestling			
Event	1st	2nd	3rd
Olympic Games	1	0	0
World Championships	6	0	0
World Junior Championship	1	0	0
World Cup	1	1	0
Other	6	1	2
Total	**15**	**2**	**2**

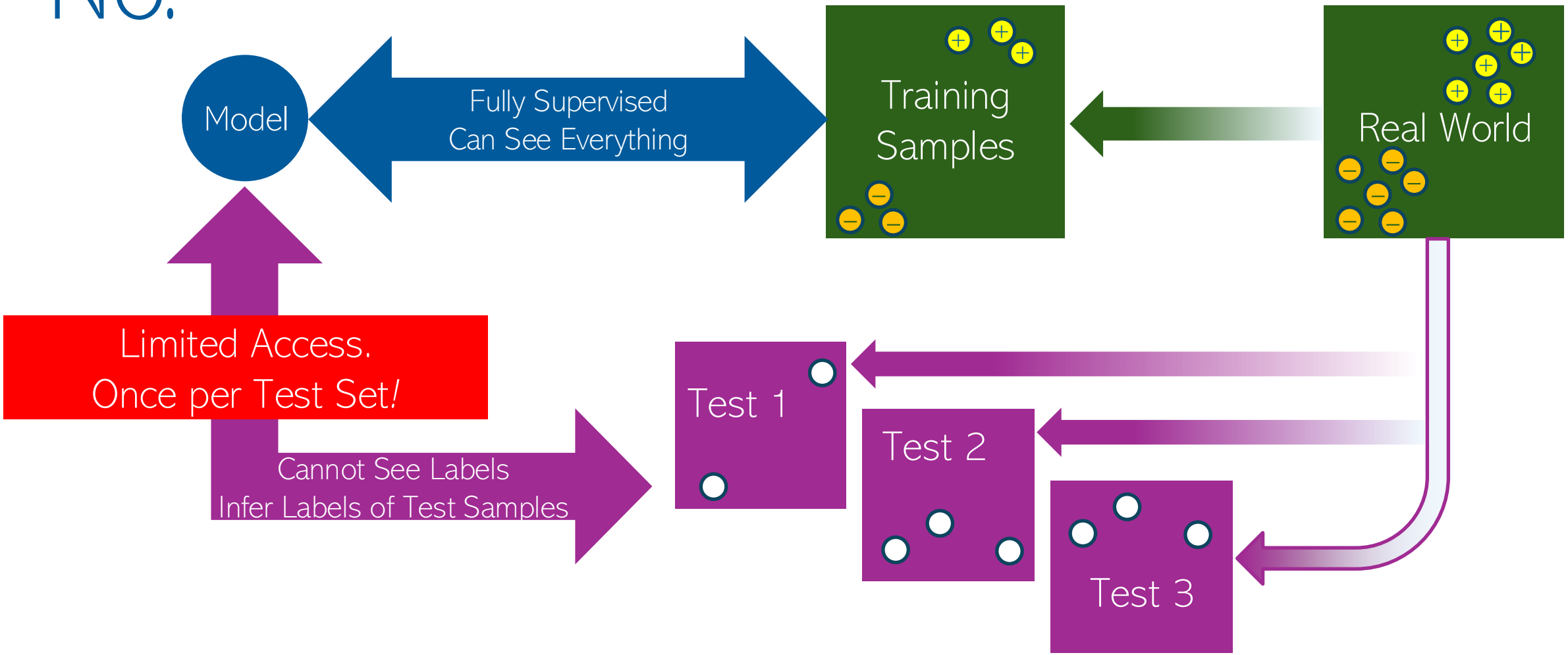
Hamid Sourian, 59KG Greco-Roman Wrestler, Iran, Rio Olympic 2016

7 - 0 ⟶ Lost to a Fall (Pin)

Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a **single test set** enough to decide on a model?

No.



Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a **single test set** enough to decide on a model?

No.

Test Sets	F_1-Measure
Midterm 1	0.4
Midterm 2	0.7
Final Exam	0.6
Average	**0.57**
Harmonic	**0.54**

Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a single test set enough to **compare models**?

No.

Test Sets	2NN-F_1-Measure	5NN-F_1-Measure
Midterm 1	1.0	0.4

2NN too good (disturbingly good) to be true …

Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a single test set enough to **compare models**?

No.

Test Sets	2NN-F_1-Measure	5NN-F_1-Measure
Midterm 1	1.0	0.4
Midterm 2	0.4	0.7
Final Exam	0.4	0.6
Average	0.6	0.57
Harmonic	0.5	**0.54**

After multiple tests, 5NN is conservatively better.

Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a single test set enough?

No, but creating multiple test sets are very costly …

design exam questions, order copies, pick up copies, 100(1+10) papers (1 Tree), missing one lecture, ask for invigilators, complains about questions, book room for marking, marking, complains about marking, review marking, …*

Unless it's a critical tasks like a missile system,

We fall back to "Yes". A single test set is sufficient*!*

Final Exam

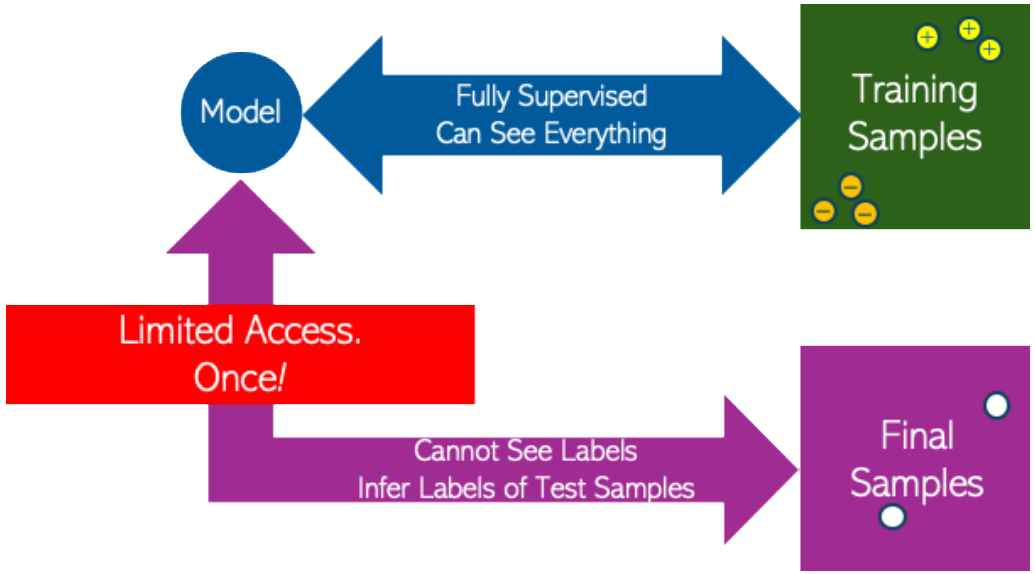
For critical tasks, if interested: Paired Significance Test, …

Evaluation Methodology: Protocol to Count Inferences

Is evaluation on a **single test set** enough to decide on a model?

Yes, but the model prepares for it by **self validations**, i.e., obtains the best performance prior to the single final test.

How?



Evaluation Methodology: Protocol to Count Inferences

Model Tuning



Having full access to the entire training set, …

lecture slides and recordings, …

Evaluation Methodology: Protocol to Count Inferences

Model Tuning

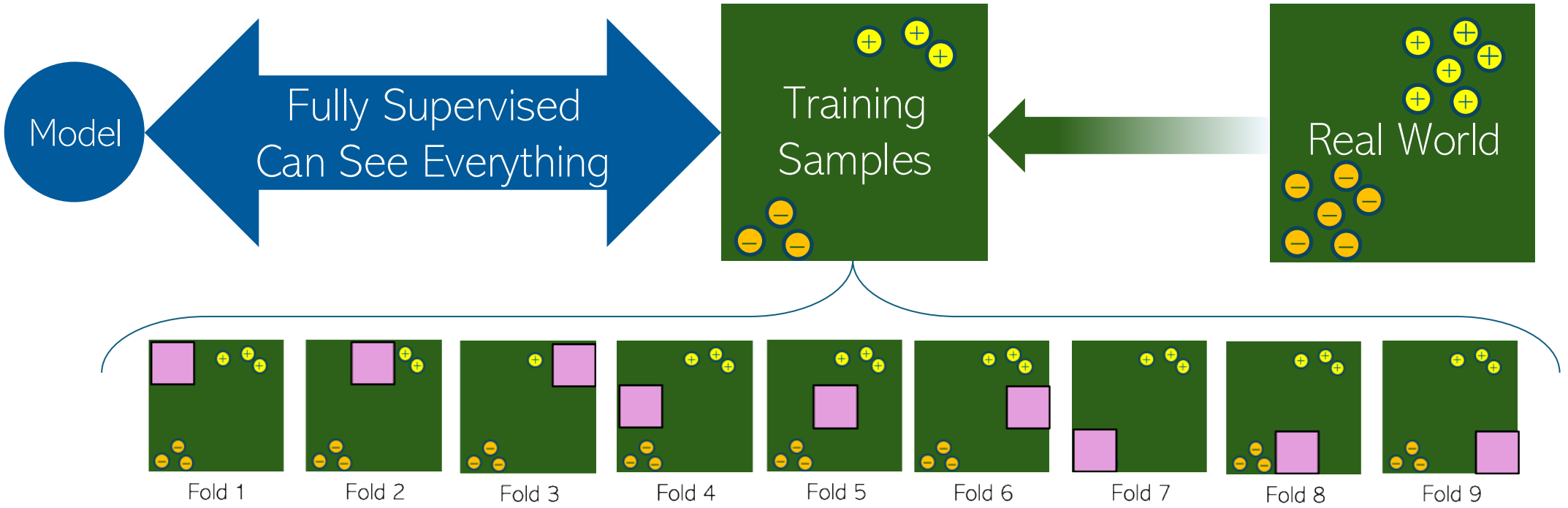


Create **internal test sets** by hiding equally sized subsets (**folds**) …

question *by hiding equal number of slides …*

Evaluation Methodology: Protocol to Count Inferences

Model Tuning

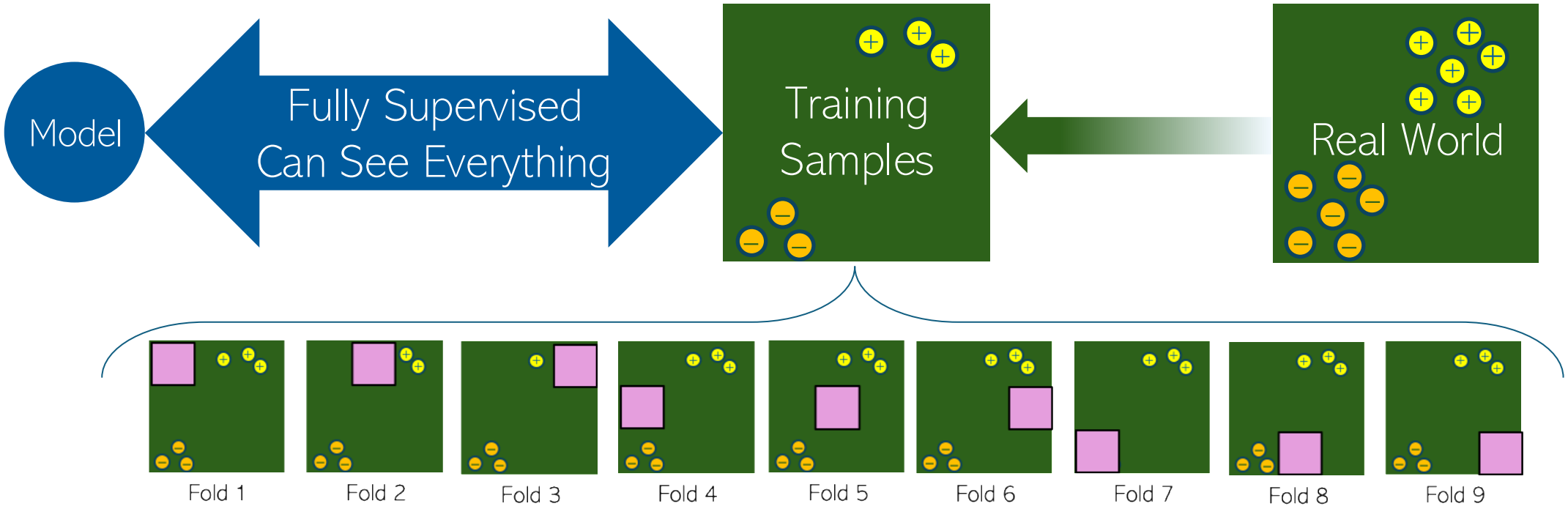


Train on **un**masked samples

Evaluate on **masked** samples, aka Validation (Development) Set.

Evaluation Methodology: Protocol to Count Inferences

Model Tuning by n-fold Cross-Validation

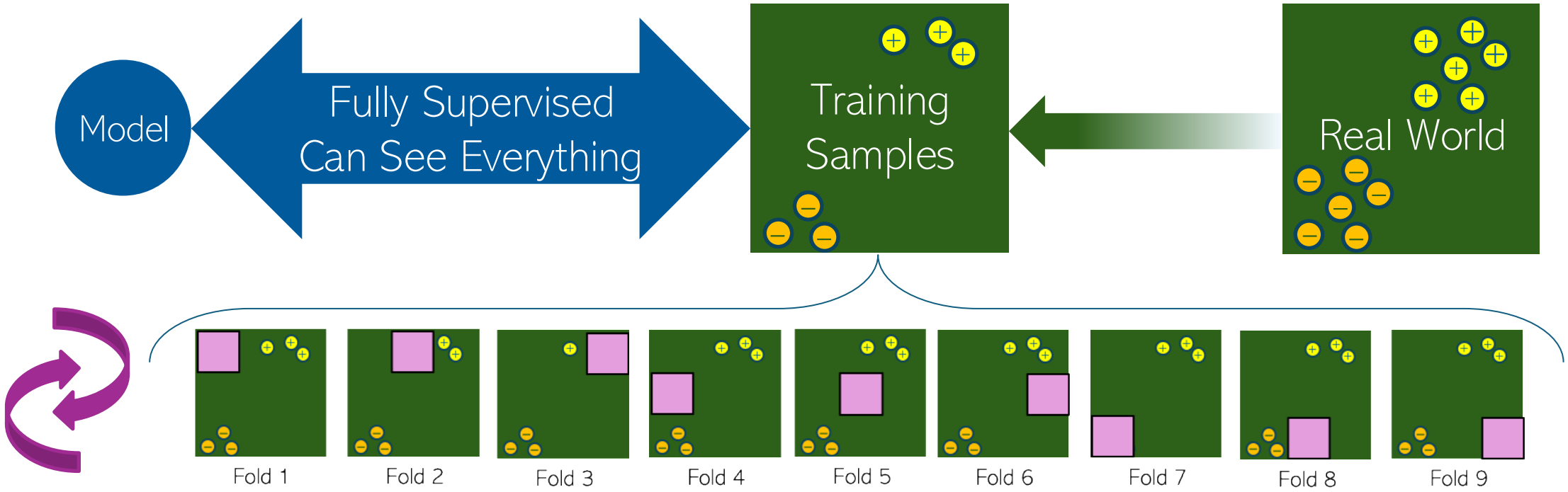


9 internal trainings, 9 internal validations

9-fold Cross Validation

Evaluation Methodology: Protocol to Count Inferences

Model Tuning by n-fold Cross-Validation



A data mining developer has **unlimited opportunity** to find issues and fix them **prior to final test**.

Evaluation Methodology: Protocol to Count Inferences

Model Tuning by n-fold Cross-Validation

Training Set	Validation Sets	2NN-F_1-Measure	5NN-F_1-Measure
Folds 2,3,4,5,6,7,8,9	Fold 1	0.13	0.87
Folds 1,2,4,5,6,7,8,9	Fold 2	0.87	0.69
Folds 1,2,4,5,6,7,8,9	Fold 3	0.42	0.94
Folds 1,2,3,5,6,7,8,9	Fold 4	0.69	0.94
Folds 1,2,3,4,6,7,8,9	Fold 5	0.05	0.87
Folds 1,2,3,4,5,7,8,9	Fold 6	0.94	0.69
Folds 1,2,3,4,5,6,8,9	Fold 7	0.31	0.79
Folds 1,2,3,4,5,6,7,9	Fold 8	0.58	0.91
Folds 1,2,3,4,5,6,7,8	Fold 9	0.76	0.76
	Average	0.53	0.83
	Harmonic	0.26	0.82
	Final Test	☐ [0.26, 0.53] Probably Fails	☐ [0.82, 0.83] Probably Pass

Evaluation Methodology: Protocol to Count Inferences

Model Tuning by n-fold Cross-Validation

Who did this for the midterm exam?
A fair expectation is no student.
Why?

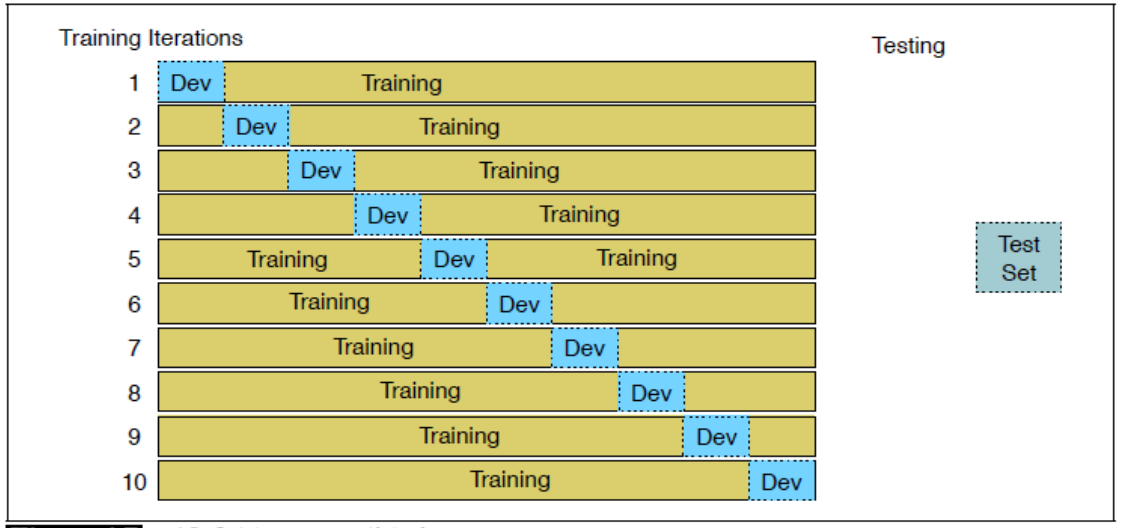


Figure 4.7 10-fold cross-validation

A fair and reasonable response:
I pay tuition. Should I also take on the entire workload of learning!?
Why doesn't the instructor run multiple tests? What are they getting paid for, then!?

Evaluation Methodology: Protocol to Count Inferences

Model Tuning by n-fold Cross-Validation

n times training

A single training usually take 1 week. For n-fold, you need n weeks!

How about kNN?

Training Set	Validation Sets	2NN-F_1-Measure	5NN-F_1-Measure
Folds 2,3,4,5,6,7,8,9	Fold 1	0.13	0.87
Folds 1,2,4,5,6,7,8,9	Fold 2	0.87	0.69
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	Average	0.53	0.83
	Harmonic	0.26	0.82
Final Test		$\simeq$[0.26, 0.53] Probably Fails	$\simeq$[0.82, 0.83] Probably Pass

39

Evaluation Methodology: Protocol to Count Inferences

Model Tuning by n-fold Cross-Validation

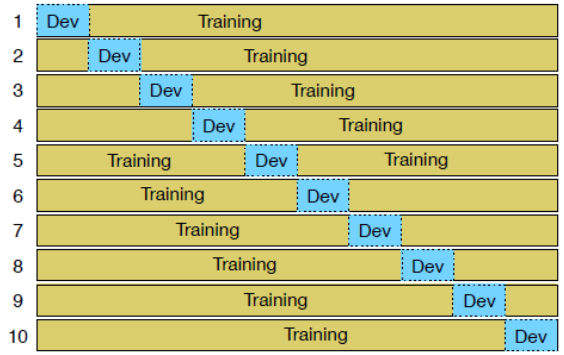
k-Nearest Neighbor does not have training. So, zero training time.

But inference is slow, n times slower. Yet test set is small. Not very costly.

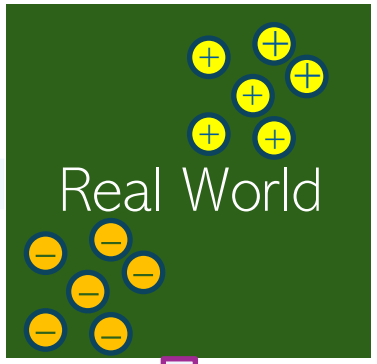
Training Set	Validation Sets	2NN-F_1-Measure	5NN-F_1-Measure
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	Average	0.53	0.83
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	Final Test	$\simeq$[0.26, 0.53] Probably Fails	$\simeq$[0.82, 0.83] Probably Pass

Evaluation Methodology: Protocol to Count Inferences

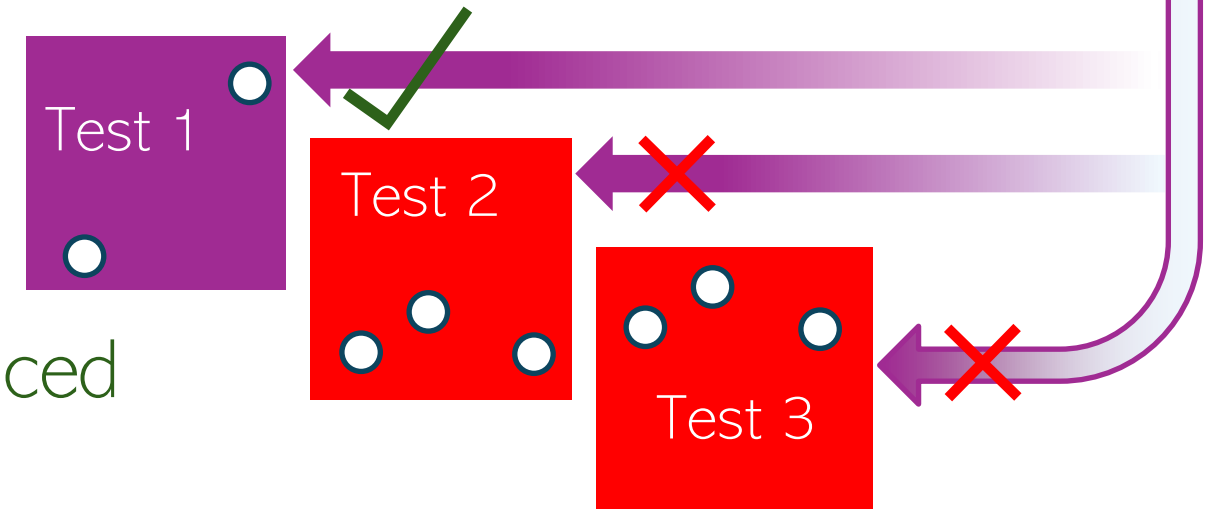
Label Distribution on Training Folds, Validation Sets, Test Set







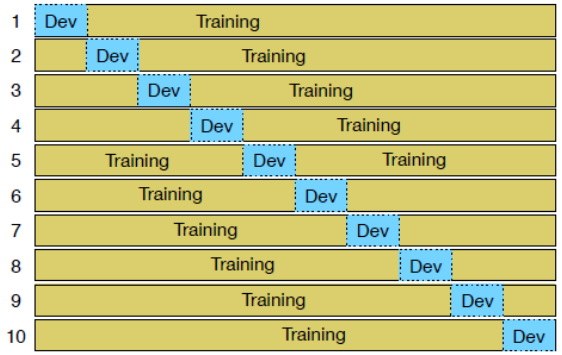
Same or like the distribution in the real world.



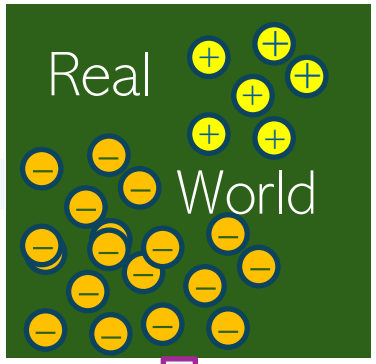
If balanced all sets be balanced

Evaluation Methodology: Protocol to Count Inferences

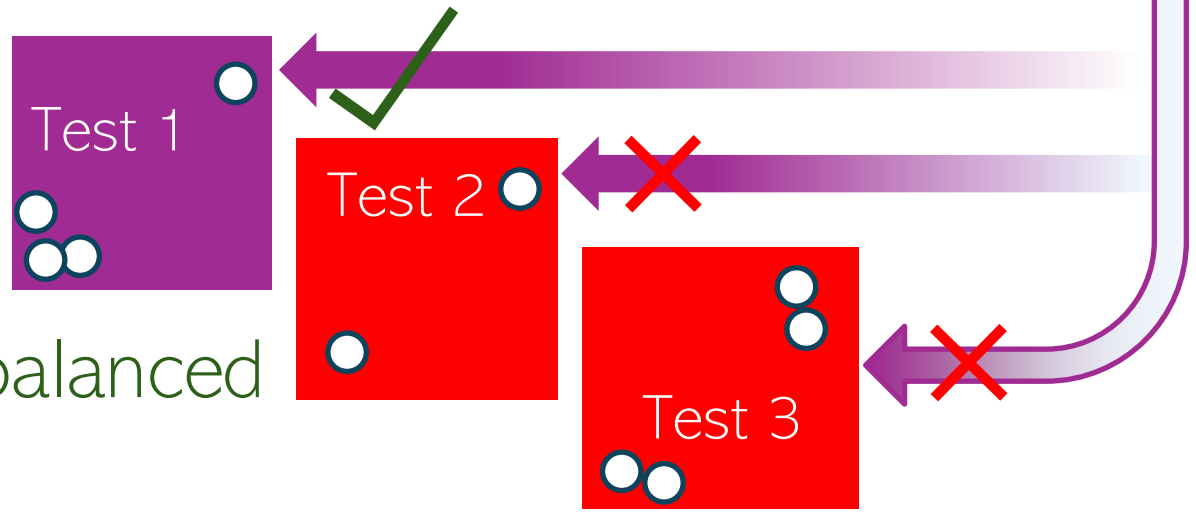
Label Distribution on Training Folds, Validation Sets, Test Set







Same or like the distribution in the real world.



If imbalanced all sets be imbalanced

Evaluation Methodology: Scikit's StratifiedKFold

Cross-Fold-Validation-kNN.ipynb

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[ ]

```python

from sklearn.model_selection import StratifiedKFold
from sklearn.neighbors import KNeighborsClassifier

skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=2026)# create 5 stratified folds

knn = KNeighborsClassifier(n_neighbors=1)# 3-NN classifier (3 nearest neighbors)

precisions = []; recalls = []; f_measures = []

for fold, (train_idx, test_idx) in enumerate(skf.split(0, y), start=1):
    0_train, 0_test = 0[train_idx], 0[test_idx]
    y_train, y_test = y[train_idx], y[test_idx]

    knn.fit(0_train, y_train)# dummy train. simply stores the training observations of all folds except current fold.

    y_pred = knn.predict(0_test) # predict labels of current fold

```

Outline: W02C-W07B

I) Task: Classification

II) Supervision: Fully Supervised

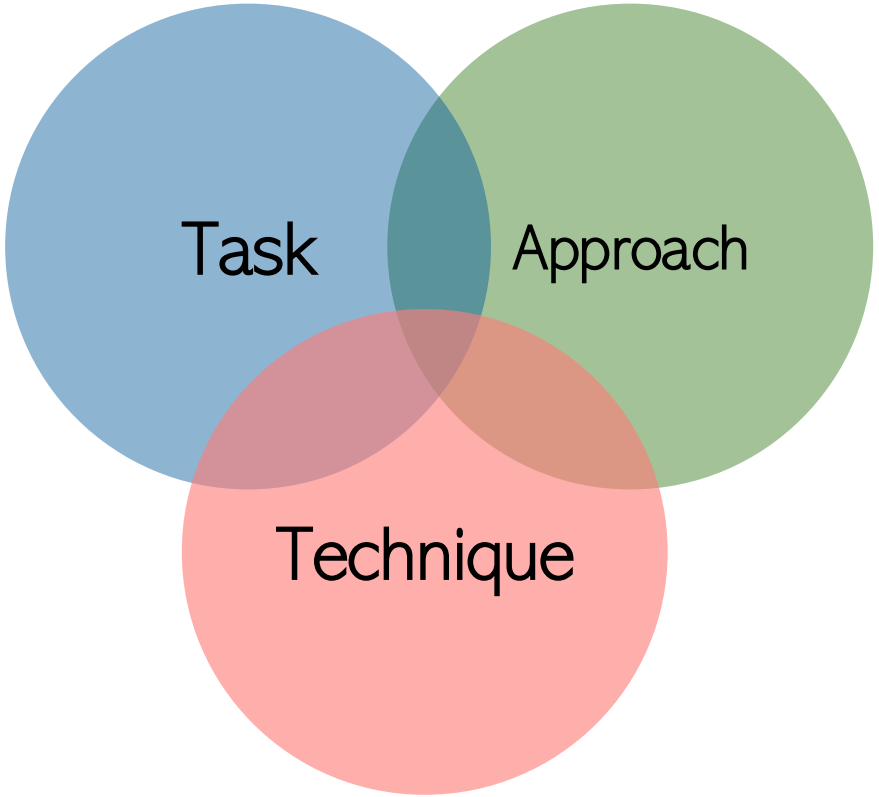
III) Technique: k-Nearest Neighbor

IIII) Evaluation

Metrics, Methodology

Outline: W07C: New Technique: Logistic Regression (LR)

Classification
Regression
Clustering
Ranking (Recommendation)
Vector Representation and Dimensionality Reduction
Language Modeling



Full/Self/Semi-**Supervised Learning**
Unsupervised Learning
Reinforcement Learning

k-Nearest Neighbor, Naive Bayes, **Logistic Regression**
Artificial Neural Networks, Graph Neural Network
Support Vector Machines
Principal Component Analysis
K-Means